25 Introduction to Bayesian Hierarchical Linear Models

We have studied a Bayesian simple linear regression model. We have also introduced some basic principles of hierarchical models. Now we'll combine elements to formulate Bayesian hierarchical linear regression models.

This handout corresponds to Chapters 15-17 of (Johnson et al. 2022). This handout only contains a brief introduction; see (Johnson et al. 2022) for a much more thorough analysis.

We'll consider the `cherry_blossom_sample` data in the `bayesrules` package which contains the `net` running times (in minutes) and `age` for participants in the annual 10-mile Cherry Blossom race held in Washington, D.C. Each runner in the sample is in their 50s or 60s.

```
library(bayesrules)

data(cherry_blossom_sample)
running <- cherry_blossom_sample |>
  select(runner, age, net) |>
  na.omit()

head(running)
```

```
# A tibble: 6 x 3
  runner   age   net
  <fct>  <int> <dbl>
1 1         53  84.0
2 1         54  74.3
3 1         55  75.2
4 1         56  74.2
5 1         57  74.2
6 2         52  82.7
```

1. Suppose we fit a Bayesian simple regression model of `net` on `age`. What do the slope and intercept parameters represent in context? What might be a reasonable prior? (You can assume that the `age` variable has been centered by subtracting the sample mean.)

2. Fit the model from the previous part (see the results in Section 17.1 of (Johnson et al. 2022)). Do the results seem reasonable? (Hint: consider the posterior distribution of the slope.)

3. Each of the runners in the sample has run the race in multiple years. What assumption of the simple linear regression model is violated?

4. Formulate a Bayesian hierarchical linear model which assumes the intercept (but not slope) varies by runner. What do the parameters represent in context?

5. Fit the model from the previous part (see the results in Section 17.2.4 of (Johnson et al. 2022)). What are some features of the analysis you would explore? Consider the posterior distribution of the slope associated with `age`; does the posterior distribution based on this hierarchical model seem more reasonable than the one based on the simple linear regression model?

6. Formulate a Bayesian hierarchical linear model which assumes both the intercept and the slope vary by runner. What do the parameters represent in context?

7. Fit the model from the previous part (see the results in Section 17.3.3 of (Johnson et al. 2022)). What are some features of the analysis you would explore?

8. Compare the two hierarchical regression models. What are some similarities? What are some differences?